

A Lightweight Pipeline Edge Detection Model Based on Heterogeneous Knowledge Distillation

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Abstract—The pipeline safety warning system (PSEW) is an important guarantee for the safe transportation of energy pipelines. Given the constraints of deploying detection models at resource-limited pipeline stations, there is a compelling need to develop efficient, lightweight models suitable for edge device applications. This brief introduces an adaptive heterogeneous model knowledge distillation network (AHKDnet) for edge deployment of pipeline network detection models. The global information and long-distance dependency relationships from the ViT-based teacher network are transferred to the CNN-based shallow student network. We introduce the learnable modulation parameters to optimize target information enhancement, reducing the impact of irrelevant information. By embedding the model selection at each stage of knowledge distillation, the performance collapse of student models caused by misleading cross-architecture knowledge is avoided, and model convergence is accelerated. Experiments on three actual scene datasets of pipeline networks show that AHKDnet outperforms the state-of-the-art KD methods and has strong generalization ability. Notably, AHKDnet enhances the recognition performance of shallow student networks by an average of 10%, highlighting its efficacy and potential for practical applications. Our method can provide a new reference for edge deployment of PSEW.

Index Terms—Pipeline network safety, knowledge distillation, heterogeneous model, lightweight, distributed optical fiber.

I. INTRODUCTION

PIPELINE transportation is one of the essential modes of energy transportation, especially for oil and gas

resources [1]. Due to the long distance of pipeline transportation, various disruptive events (such as mechanical excavation, construction, human activities, etc.) may occur in an open environment, posing a threat to the safety of pipelines. In 2022, the Nord Stream natural gas pipeline suffered external damage and leakage, resulting in multiple explosions. The oil pipeline in Kansas, USA, leaked over 14000 barrels of crude oil, causing significant economic losses and environmental pollution. Thus, it is beneficial to conduct intrusion detection around pipelines to promptly recognize such hazardous events and implement corresponding measures. In recent years, the pipeline safety early warning (PSEW) system based on distributed optical fiber sensing (DOFS) technology has become an important technology for preventing pipeline damage and ensuring energy transportation [2]. It employs communication optical cables along the pipeline as a sensor array, recognizing signal types through signal processing and Artificial Intelligence (AI) algorithms. Subsequently, early warning and verification of hazardous events are conducted. DOFS has extensive application prospects in energy, industry and other fields due to its continuous monitoring, high resolution, and low cost.

The key to the PSEW system is the fast and accurate intrusion detection algorithm. Therefore, effective and reliable feature extraction and recognition network for damage events comes to play and has received notable attention [3]. In previous studies [4], [5], threshold-based and statistical machine learning methods were used for signal processing and intrusion event recognition in the single environment. At present, deep learning methods have become the mainstream method of recognition algorithms to address the impact of more complex environments and background noise in the field of DOFS [6], [7]. Yang et al. proposed a real-time action recognition method for long-distance PSEW systems [1]. The system can quickly identify and locate third-party damage events by calculating two complementary features and constructing a novel action recognition deep learning network. Wu et al. proposed a two-level multi-task learning (MTL) enhanced smart fiber-optical distributed acoustic sensing (sDAS) system to simultaneously realize ground event recognition and localization [8]. These studies fully confirm that the combination of new intelligent sensors and AI effectively improves the intelligence and safe operation of pipelines [9], [10].

However, there are still some challenges in the actual deployment of pipeline networks. It is usually necessary to deploy multiple devices in stations or valve chambers along the

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pipeline to achieve comprehensive coverage of monitoring due to the long transportation distance of pipelines. Stations often have limited computing power and storage resources, which affect the size and processing speed of deep learning models. Currently, many studies [11], [12] aim to test deep models on a single device without considering the global deployment of pipeline networks. Our previous research [13], [14] also found that the performance of using vision transformer (ViT) structural models for classification after image encoding of time series signals is better than that of CNN-based structural models, which can improve the recognition performance of PSEW systems. Nevertheless, models based on ViT structure require larger data and more parameters, while CNN structure models have relatively lower computational complexity. Consequently, it is necessary to compress model to balance the performance and time cost in pipeline network applications.

To achieve the goal of deploying efficient lightweight models at the edge, we propose an adaptive heterogeneous model knowledge distillation network architecture, AHKDnet that simultaneously uses a responses-based and features-based distillation method for model learning in logits space. To combine the advantages of ViT structure and CNN structure, the ViT-based model is regarded as the teacher network, while the CNN-based model serves as the student network, enabling the student model to learn the global information and long-distance dependency relationships extracted by the teacher model. This also makes the CNN model strengthen the learning of position information, compensating for the spatial information loss during convolution and pooling processes. Therefore, it is necessary to project the features extracted from heterogeneous models into a unified spatial alignment for knowledge distillation (KD). Moreover, we add learnable modulation parameter and model selection at each stage of KD to ensure the effectiveness of knowledge transfer. Enhancing target information can help improve the learning effectiveness of student models on teacher models, and alleviate the impact of target class soft labels when teacher models provide under prediction. Model selection can accelerate the convergence speed of student models by judging the using/not using KD. Our major contributions are summarized as follows:

- 1) We propose an adaptive heterogeneous knowledge distillation network (AHKDnet) for limited edge deployment of pipeline intrusion detection models. Compared with the deep learning framework without KD, it can not only improve the recognition accuracy of student model, but also reduce the actual deployment requirements.
- 2) Considering the differences of target class distribution in heterogeneous models, we have improved the modulation parameters that can learn to optimize target information enhancement in the loss function, further reducing the impact of irrelevant information in logits space.
- 3) We introduce the mechanism of model selection based on threshold in the feature distillation of intermediate layers. It can limit the performance collapse caused by misleading cross-architecture knowledge, thus optimizing the effectiveness of KD in each stage of the model.

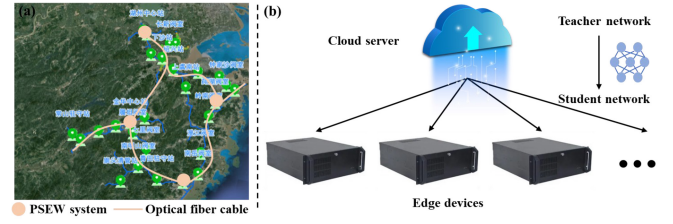


Fig. 1. (a) The diagram of PSEW deployed in the pipeline network, (b) The deployment strategy of edge detection models in pipeline networks.

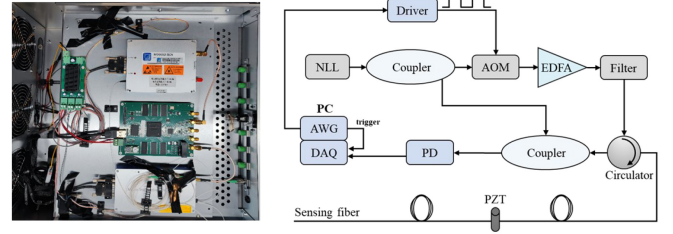


Fig. 2. The structure diagram of DAS system.

The remainder of this brief is as follows. Section II introduces the structure of PSEW system and its application in smart pipeline network. Section III provides a detailed explanation to AHKDnet. Section IV shows the experimental results to demonstrate the superior performance of the proposed model. In Section V, this brief is summarized.

II. PSEW SYSTEM FOR SMART PIPELINE NETWORK

PSEW system is an important component of smart pipeline network, which can achieve intelligent perception and dynamic monitoring of pipelines. At present, PSEW mainly consists of distributed optical fiber acoustic sensing (DAS) system and unmanned aerial vehicle inspection system. Among them, the DAS system utilizes the high sensitivity of coherent Rayleigh scattering in optical fiber, which can perceive environmental vibration and acoustic field at long distances with high spatiotemporal resolution. However, the monitoring range of a single device is limited, usually within 50km.

As shown in Fig. 1 (a), multiple devices need to be installed in the station or valve chamber of the pipeline network to achieve continuous monitoring of the pipeline. The DAS system is connected to the sensing fibers laid parallel to the pipeline, continuously monitoring the external acoustic signals of the pipeline. Fig. 1 (b) shows the deployment strategy of our lightweight edge detection model for cloud-edge collaborative. By training a powerful teacher network in the cloud for KD, the performance of the student network deployed by edge devices can be improved.

The structure of DAS system is shown in Fig. 2. The system uses narrow linewidth laser (NLL). The beam is modulated into narrow-band pulsed light by an acousto-optic modulator (AOM). The pulsed light is amplified by an erbium-doped fiber amplifier (EDFA) and filtered. The modulated pulsed light enters the sensing fiber through the coupler and circulator. The sensing fiber is disturbed by piezoelectric ceramics (PZT) to generate Rayleigh backscattered light signal. The optical

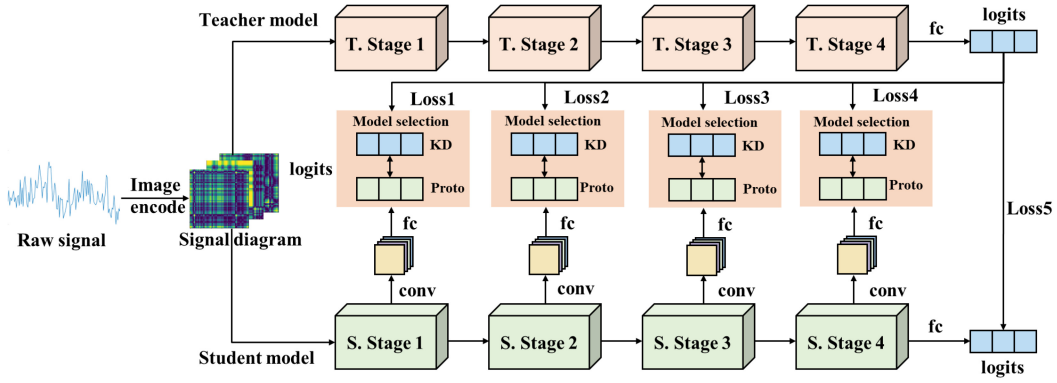


Fig. 3. The architecture of AHKDnet.

signal is converted by photodetector (PD) and triggered by an arbitrary waveform generator (AWG) to collect the signal on the data acquisition (DAQ). Finally, the signal is processed through the warning model deployed on the upper computer.

III. METHODOLOGY

Knowledge distillation (KD), as a model compression method, has broad application prospects in industry [15]. KD optimizes the student model by minimizing the difference in output between the teacher model and the student model, typically using soft labels as the training objective for the student model. KD aims to extract knowledge from higher accuracy teacher networks to improve the performance of lightweight student networks [16]. To integrate the advantages of different architecture models, we propose the AHKDnet method based on the OFA-KD framework [17] to improve the performance of heterogeneous model knowledge distillation in pipeline network scenario. OFA-KD projects intermediate features into an aligned latent space, causing specific architecture related information to be discarded. It significantly improves the distillation performance between heterogeneous system structures. Fig. 1 illustrates the framework of AHKDnet. Gramian angle field (GAF) is applied to transform the 30s DAS raw signal into 2D image as model input [13]. The structure of the model consists of heterogeneous KD mechanism, the optimization target enhancement with learnable parameters and the feature distillation of intermediate layers with model selection.

A. Heterogeneous Knowledge Distillation Mechanism

The ViT-based models have more advantages in event classification of complex environmental signals, while the CNN-based models have relatively lower computational complexity. The ViT-based models can simultaneously consider both global and local feature extraction in different layers due to the role of self-attention mechanism. The CNN-based models follow a feature extraction process from local to global, making it difficult for shallow networks to retain more refined local spatial information. Previous studies have demonstrated the feature mismatch caused by representation differences between heterogeneous models by calculating their centered kernel alignment (CKA) [17]. CKA is a feature similarity

measurement allowing cross-architecture comparison achievable, as it can work with inputs having different dimensions. To achieve heterogeneous model KD of intermediate layer features, the intermediate layer features are transferred to an aligned logits space and matched with the final output (logits) of the teacher network.

$$\text{CKA}(K, L) = \frac{D_{\text{HSIC}}(K, L)}{\sqrt{D_{\text{HSIC}}(K, K) D_{\text{HSIC}}(L, L)}} \quad (1)$$

$$D_{\text{HSIC}}(K, L) = \frac{1}{(n-1)^2} \text{tr}(KHLH) \quad (2)$$

where D_{HSIC} denotes the Hilbert-Schmidt independence criterion. K and L represent Gram matrices of the features. H is the centering matrix. The $\text{tr}()$ represents the trace of a matrix.

B. Learnable Optimization Target Enhancement

Heterogeneous models have inherent differences in learning abilities and preferences due to their different inductive biases [17]. To enhance the information of target class, we added a learnable parameter related to the target class to participate in learning optimization at each stage, achieving adaptive enhancement of the target information. The difference from fixed modulation parameters is that it updates the values of each stage based on gradient contributions. The amount of information affected by the teacher network varies due to the changes in the receptive field of the CNN model at different stages. There are differences in the optimal fixed parameters for different teacher models, and the most suitable value for each stage is also different. The loss function is defined as.

$$\text{Loss}_{\text{ahkd}} = -(1 + p_c^t)^{\alpha \hat{\gamma}} \log p_c^s - \lambda E[p_c^t, \log p_c^s] \quad (3)$$

where p_c^t and p_c^s represent the prediction probability of the predicted class for the teacher model and the student model, respectively. p_c^t and p_c^s are the probability of the target class for the student model and the teacher model. α is used to control the learning rate of $\hat{\gamma}$. $\hat{\gamma}$ denotes learnable modulating parameter to achieve learnable optimization of target information enhancement. α and $\hat{\gamma}$ adjust the differences in learning ability and preferences of heterogeneous models based on the probability of target class soft labels provided by the teacher model. λ is a hyperparameter controls the trade-off between one-hot label and soft label.

TABLE I
COMPREHENSIVE COMPARISON RESULTS OF DIFFERENT MODELS AND DISTILLERS

Teacher	#Params	FLOPs	Student	#Params	FLOPs	Original model			Distiller				
						T.	S.	KD	RKD	DKD	DIST	OFA	Ours
ViT-B	86M	55.4G	ResNet18	11.7M	1.8G	91.48	76.34	78.69	75.98	82.32	77.42	85.55	85.59
DeiT-B	86M	17.5G	ResNet18	11.7M	1.8G	93.21	76.34	79.73	76.12	81.92	78.55	85.95	86.47
Swin-B	88M	15.4G	ResNet18	11.7M	1.8G	95.68	76.34	80.16	76.68	82.84	78.86	86.15	86.85
ViT-B	86M	55.4G	EfficientNet-B0	5.3M	0.39G	91.48	77.69	79.75	77.42	86.42	78.31	86.67	87.28
DeiT-B	86M	17.5G	EfficientNet-B0	5.3M	0.39G	93.21	77.69	80.42	77.63	86.90	77.96	87.12	88.16
Swin-B	88M	15.4G	EfficientNet-B0	5.3M	0.39G	95.68	77.69	81.03	77.72	87.59	78.48	88.53	89.48

C. Adaptive Model Selection With KD

We introduce a model selection mechanism to ensure the gain effect of KD [18]. Specifically, the effect with KD on model convergence can be determined by adding the threshold to optimize the update speed of gradients in the model. We train an additional local model in parallel at each stage, which update gradients through heterogeneous model distillation and update gradients without KD learning, respectively. By calculating the gain of gradient descent on model convergence, the stage model with a greater contribution is selected to use the next round of training using the model. Thus, it suppresses the performance collapse of student models caused by misleading heterogeneous knowledge and improve the convergence performance of student model.

$$G_j = \|g_{kd}^j\|^2 + 2[g_i^j]^\top g_{kd}^j \quad (4)$$

$$\begin{cases} G_j \geq \xi_j, & \text{the model with KD;} \\ G_j < \xi_j, & \text{the model without KD.} \end{cases} \quad (5)$$

where G_j is the performance gain by using the KD method at stage j . g_{kd}^j and g_i^j represent gradients with and without KD at stage j , respectively. ξ_j denotes the threshold of the model contribution at stage j . If $G_j \geq \xi_j$, the sum of the step size of gradient has a positive contribution to the local model update. Then, the local model for the next training round should use KD. On the contrary, there is no need to learn through KD.

IV. EXPERIMENTS

The experimental data is collected from DOFS signals in actual scenes of natural gas pipelines in Zhejiang Province, including three kinds of DOFS datasets with different sizes, types and scenes. We select three types of events on each pipeline, including intrusion events, interference events, and environmental noise. Each type has 1000 samples, totaling 9000 samples. The system sampling frequency is 4Hz, and the sample signal length is 30s, with a total of 120 sampling points. In paper [14], a more detailed introduction is provided to the datasets. We convert the original signals into 2D images by the image encoding method of Gramian Angular Field (GAF), and unify the resolution of images to 224x224. It can preserve the complete information of the signal and maintain its dependence on time. The CNN-based models are trained using the SGD optimizer with an initialized learning rate of 0.05, while other ViT-based models are trained using the Adam optimizer (set $\beta_1 = 0.9$, $\beta_2 = 0.99$). All experiments are trained for 100 epochs using 5-fold cross validation.

AHKDnet is compared with previous KD methods, including KD [19], RKD [20], DKD [21], DIST [22], and OFA [17]. We employ ViT-based teacher networks, including ViT-B [23], Swin-B [24], and PipelineADVinT [14], as well as CNN-based shallow student networks, ResNet18 [25] and EfficientNet-B0 [26]. Moreover, we compared the performance improvement of student networks on two directions of heterogeneous knowledge distillation, from ViT to CNN and from CNN to ViT. The performance improvement of the proposed method was experimentally validated on three datasets. The accuracy (ACC) is used as an indicator to evaluate the performance of each model. The higher the accuracy, the stronger the recognition performance of the model, as shown below.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where TP , FP , FN , TN represent true positives, false positive, false negative and true negative respectively.

The comparative experimental results between the proposed method and previous work indicate that AHKDnet has superior performance in Table I. Overall, it can improve the accuracy of the original student model by approximately 10%. When the performance of teacher and student networks is enhanced, the distillation effect is also better. PipelineADVinT/EfficientNet-B0 improves by 11.79% compared to the original student model, reaching 89.48%. However, the improvement effect is limited compared to the improved OFA model, with the accuracy increase of approximately 1%. After checking the misclassified samples, it was found that improvements in the OFA model can only be effective when special samples are prone to misleading. This may play a crucial role for practical applications in real-time signal recognition. Moreover, it can be seen from the number of parameter and FLOPs that the proposed method can significantly improve the recognition ability of student models with small parameter quantities, while reducing computational costs.

Fig. 4 (a) indicates that ViT-based is more suitable as the teacher network in pipeline scenarios. Compared with CNN-based teacher networks, it can significantly improve the performance of trained student networks. On the one hand, it is due to the limited performance of shallow ViT models as student networks. On the other hand, the global information obtained by the ViT model of attention mechanism can be transmitted to the CNN model to compensate for the missing feature information during the convolution process. As shown in Fig. 4 (b), the test results on three datasets show that the proposed method can improve the average

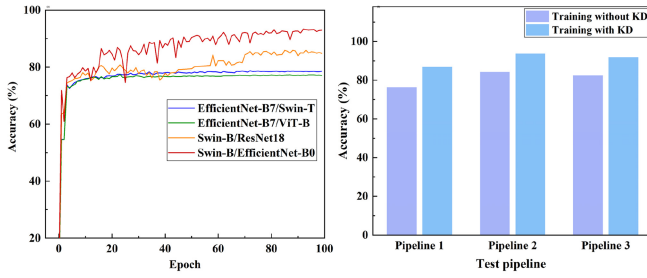


Fig. 4. (a) Test accuracy curves in different KD directions, (b) Experimental comparison of the proposed methods on different pipelines.

recognition accuracy of the model by 10%, which has strong generalization ability and stable improvement on lightweight student networks.

V. CONCLUSION

This brief proposed an adaptive heterogeneous model knowledge distillation network for the pipeline network deployed detection models at the edge. The proposed method effectively improves the recognition performance of lightweight CNN-based models by projecting logits space for heterogeneous model knowledge distillation. We introduce learnable modulation parameter and model selection at each stage of knowledge distillation for target information enhancement and the effectiveness of knowledge transfer. Experiments on several pipelines have confirmed the superiority of the proposed method. Experimental results show that the accuracy of AHKDnet is 10% higher than that of the original student model, and about 1% higher than that of SOTA KD method.

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